

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

Q.23 Explain the construction details and operation of JFET. What are the differences between JFET and MOSFET?

Q.24 Draw the circuit diagram of AND and NOR gate. Explain why NOR and NAND gates are known as universal gate. Draw OR gate using NAND gate.

Q.25 a) Write any four applications of PN junction diode.

b) Perform the following operations-

i) $1011+1011$

ii) $1110-0101$

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Roll No.

**2nd Sem / Instrumentation & Control engg.,
Medical electronics**

Subject : Analogue and Digital Electronics

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

Q.1 If the doping concentration of any diode is $1:10^6$, then the diode is

a) Zener

b) normal PN diode

c) LED

d) None of the following

Q.2 The ripple factor of a full wave rectifier is approx

a) 1.21

b) 1.38

c) 0.48

d) 0.90

Q.3 The octal equivalent of a binary number $(10110011)_2$ is

a) $(236)_8$

b) $(867)_8$

c) $(632)_8$

d) $(263)_8$

Q.4 Which of the following option is correct for the two input (A and B) NAND gate

- a) If $A=1$ and $B=1$ then output $=1$
- b) If $A=0$ and $B=0$ then Output $=0$
- c) If $A=0$ and $B=1$ then Output $=1$
- d) If $A=1$ and $B=0$ then Output $=0$

Q.5 A latch circuit with a clock input is known as

- a) Shift register b) Flip-Flop
- c) Adder d) A/D Converter

Q.6 For Ge diode, the value of cut-in voltage is

- a) 1V b) 0.3 V
- c) 0.7 V d) 1.08 V

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

Q.7 Expand JFET.

Q.8 A NOR gate is the combination of NOT and OR gate.(T/F)

Q.9 Number of select lines required to design a 8:1 MUX are _____.

Q.10 The depletion region decreases if the reverse bias voltage across the normal P-N diode increases.(T/F)

Q.11 What is the ripple factor of rectifier?

Q.12 The material whose conductivity lies between insulator and conductor is known as _____.

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

Q.13 Explain A/D and D/A converter.

Q.14 Draw the 8:1 MUX circuit and also write the truth table.

Q.15 What are the differences between Normal P-N diode zener diode.

Q.16 Explain the working of a zener diode with diagram.

Q.17 Write the symbol and truth table of XNOR gate.

Q.18 Write down any four applications of MOSFET.

Q.19 Explain the CC configuration of BJT with diagram.

Q.20 Convert the following -

- a) $(7A)_{16} = (\dots\dots\dots)_2$
- b) $(101101100101)_2 = (\dots\dots\dots)_8$

Q.21 Write any four difference between half wave and full wave rectifier.

Q.22 Draw the full wave center tapped rectifier circuit and also calculate the rectification efficiency of it.